Grading Machine-Originated Plant Writes: Separately Inspectable Evidence for Industrial Agent Teams

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***Abstract*—Large language model (LLM) agents increasingly generate industrial control changes, yet the safeguards surrounding each proposal—write authority, active limits, human approval, device acceptance, process improvement—are rebuilt per deployment inside application code. This paper presents AgentWorkShop, a governed supervisory execution layer that turns these concerns into a single path for agent teams: every intervention is one governed intervention whose admission, approval, actuation, graded write acknowledgment, verdict, and compensation remain separately inspectable evidence states. Team identities reach semantic data-acquisition (DAQ) and write-control (DCW) nodes through explicit capabilities under a fail-safe line master switch; drivers, versioned production objects, plugin points, and a typed software development kit (SDK) form the integration surface, with process models, objectives, and policies external. The evaluation exercises this path through protocol-closure and mechanism-ablation drills, agent-team missions under scripted policies, a deterministic controller loop, and four concurrent process missions, then stresses it with a deployment-scale regression campaign on one live multi-line instance. Across seven benchmark executions on the committed harness, all 600 logged checks pass. The evidence grading is falsifiable and separated from admission: under success-flag semantics an archived publish-only failure receipt reads success, and removing the admission window interlock drops probe interception from 18/18 to 12/18 while the hard range holds in every arm. The evaluation is simulation-tier; no optimization-benefit claim is made.**

***Index Terms*—Agent teams, human-in-the-loop, industrial informatics, LLM agents, supervisory control, write governance.**

I. Introduction

LARGE language model (LLM) agents are moving from advising industrial operations to acting on them: recent systems generate control updates for industrial cyber-physical systems from multi-modal input [1], pair each factory module with an LLM agent that dispatches skills over a tool protocol [2], explore resource capabilities in multi-agent manufacturing [3], coordinate shop-floor tasks [4], and have been studied as real-time controllers of a robot production line [5]. What these demonstrations leave implicit is everything that surrounds the proposal: who may change which parameter, which limits are active for the current product and recipe, whether a human approved, whether the device accepted the command, and whether the process improved.

Because these concerns are normally embedded in application code, they are rebuilt per process, protocol, and agent framework—duplicated drivers, ad-hoc permission checks, logs untied to decisions—and a task state such as COMPLETED can coexist with a missed process objective. The four scenario plants (simulated in this paper) differ—injection molding, wastewater treatment (WWTP), annealing, and BOPET (biaxially oriented polyethylene terephthalate) film—but all need an auditable bridge from a team-level task to heterogeneous acquisition and control nodes. Industrial LLM work to date, by contrast, is mostly generation, diagnosis, and single-domain control (e.g., [6]).

AgentWorkShop addresses this execution gap through member–node bindings (a member being an engine-backed channel identity) and shared line, product, recipe, and run context. The response is organizational rather than algorithmic: each intervention is one governed transaction whose admission, approval, actuation, graded acknowledgment, three-way setpoint consistency, verdict, and compensation are left as separate evidence records. Nodes expose engineering quantities rather than register addresses; one governed member–node binding path serves every client, where per-application integration would duplicate drivers, permissions, and logging. Fast regulation, certified protection, and the choice of optimization algorithm remain outside the framework.

The paper makes four contributions:

1) a **capability-based industrial execution architecture** connecting agent-team identity, semantic data-acquisition (DAQ) and write-control (DCW) nodes, heterogeneous protocol drivers, and line–product–recipe–run context, with task membership distinct from physical-write authority;
2) a **reusable integration boundary** of self-describing node and driver contracts across six protocol families (five exercised in the evaluation), versioned production objects, plugin extension points, and a typed software development kit (SDK), into which the reference adapter projects scenario blueprints while process models and policies stay external (a second preset is commissioned by configuration alone, Section V-C);
3) an **evidence-preserving governed intervention workflow** separating admission, optional approval, actuation, three-level write acknowledgment with post-write verification, observation, the caller's verdict, compensation, and backstop actions—with the grad-

TABLE I
Closest LLM-based industrial control systems versus AgentWorkShop (venue, LLM role, write path, production-scoped authority, write evidence).

System (venue)	Role of the LLM	Plant write path	Authority & production context	Write evidence & verification
MADE, ICCAD 2026 [1]	multi-modal control-update generation	generated commands + safety layer	HITL verification of outputs; task-level checks	task success over scripted evaluations
Köhle et al., ETFA 2026 [2]	per-module agents, MCP skill dispatch	skill invocation via OPC UA/MQTT	module-scoped skills	task-resolution rate (6-module plant)
Lim & Kovalenko, CASE 2025 [3]	capability exploration across agents	architecture-level	resource-matching scope	architecture evaluation
Zhao et al., AEI 2026 [4]	shop-floor task dispatch	workshop experiments	coordination layer	dispatch/execution performance
Waseem et al., Mach. Learn. Eng. 2026 [5]	real-time controller in the loop	direct loop actuation (robot line)	prompt-level constraints	line KPIs under LLM control
AgentWorkShop	**replaceable strategy slot**	**governed write path: admission, HITL, dispatch, readback**	**member–node bindings; line master switch; product/recipe-scoped limits**	**graded acknowledgment; three-way consistency; attributed journal; compensation; backstop**

ing rule stated falsifiably and exercised against archived failure receipts; and

4) a **layered evaluation** exercising one governed path through protocol closure with mechanism ablation, agent-team missions, a deterministic controller loop, concurrent process missions, and a deployment-scale regression campaign on a live multi-line instance (Section V), task state and attainment reported separately in every mission report.

Scope (revisited in Section VI): real protocol stacks execute against author-implemented simulators, and scripted policies drive the governed tools (two live engine turns reported separately); fast regulation, certified protection, physical-plant transfer, and optimizer ranking are outside the claims.

II. Related Work

A. LLM Agents in Industrial Control: From Generation to Governance

Agent-based manufacturing and supervision have long linked organizational roles to plant capabilities [7], [8]; LLM-oriented industrial work now spans PLC generation and verification [6]. The closest systems, characterized in Table I, target the control action itself: MADE couples multi-modal perception to control generation behind a safety layer, scoring its own task success with no persistent production-scoped write-evidence path reported [1]; Köhle et al. dispatch module-scoped skills over a six-module plant [2]; Waseem et al. place pretrained LLMs inside the control loop as real-time controllers [5]—a layer below AgentWorkShop, so the roles are complementary; Lim and Kovalenko [3] and Zhao et al. [4] allocate work among agents, whereas this layer governs how an allocated proposal reaches the plant and what evidence it leaves. A recent review frames agentic AI's industrial value as safety-aware, explainable, HITL decision support rather than autonomous control authority [9]; the contribution here is the mechanism set that makes that positioning concrete, and Table I characterizes each system as reported in its publication.

B. Frameworks, Standards, Run-Time Guarantees, and Automation Practice

Multi-agent frameworks coordinate tasks and tools, and the Model Context Protocol (MCP) standardizes tool and inter-agent interfaces [10]; the tool it exposes, however, is a bare function signature carrying no line, product, or recipe context, with no record separating admission from execution. ISA-88 and ISA-95 supply complementary models for recipes and operations [11], [12]; AgentWorkShop's versioned recipes and run dispatch instantiate a reduced subset of that stack, and the governed life cycle transposes run-to-run control disciplines [13] onto a supervisory write path—with an open-ended proposal space, approval-mediated bindings, and compensation records where R2R has a fitted controller. A separate line enforces run-time guarantees on LLM agents—judge and guard agents, rule-based enforcement [14]; they gate software tool calls, not plant writes, and their policies could compile onto the enforcement substrate contributed here.

Permissions, range checks, approvals, and audit trails long existed in distributed-control-system (DCS) and SCADA practice—versioned batch recipes and electronic batch journals under 21 CFR Part 11 and GAMP 5 [15], [16], second-channel writeback in NAMUR open architecture [17], and setpoint adaptation in run-to-run control—and IEC 62443 codifies their security counterpart [18], while formal shielding, control-barrier functions, and simplex run-time assurance offer stronger specified properties [19]–[21]. Against that practice the difference is the proposer: a machine-generated agent identity rather than a presupposed human operator, so admission binds to member authority and prompt content stays inert, re-evaluated per request against live product and recipe state through versioned objects rather than at role-assignment time; the governed path spans four exercised write protocol families; and the evidence states—graded acknowledgment, verdict, compensation—survive strategy replacement, where the three-way check is a supervised special case of classical data reconciliation and gross-error detection [22]. Neither the five systems above, nor the review, nor a classical chain provides that combination in this form. AgentWorkShop implements source-dependent checks and a rule-based

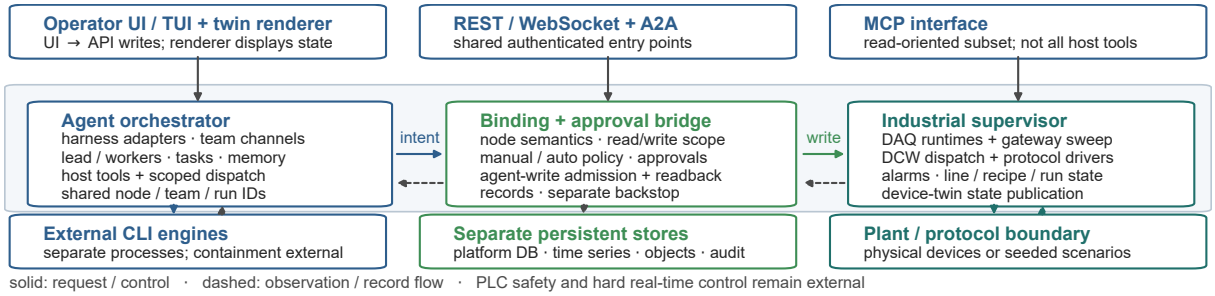


Fig. 1. System architecture and execution boundaries: platform services around shared stores, operator interfaces above, and agent engines, data stores, protocol endpoints, and the plant kept separate. Shared identifiers provide attribution but do not make storage and actuation transactional.

backstop and makes no formal protection guarantee, the backstop (Section IV-D) being a rule-based recovery action in the spirit of ISA-18.2 [23]; human oversight likewise requires more than an approval dialog [24]. Agent benchmarks and prompt-injection studies [25], [26] inform adjacent evaluations; ours separates source checks, protocol execution, mechanism ablation, controller trials, and multi-scenario missions. In sum: coordination frameworks supply no production context, standards no execution path, guard agents gate software tools, and nearest systems stop at the control action. Relative to regulated batch and DCS practice—where the operator is presupposed—the increment is narrow but specific: admitting a machine-generated proposer into that governed, plant-facing write path, with each write's evidence graded and attributed to a machine actor rather than reviewed post hoc.

III. System Architecture

AgentWorkShop fills this slot: it links team coordination to plant interaction through industrial nodes and production context (Fig. 1). Responsibilities group into five planes—logical groupings, not deployable units: a team plane owns member identity, task state, and episodic memory; an integration plane owns protocol acceptance, readback, freshness, and transport errors, presenting devices as semantic nodes whose stream feeds both the agent and a live plant-mirror view; a governance plane owns admission, approval, actuation, acknowledgment grading, verdict, compensation, and rollback; an operations plane attaches line, product, recipe, and run context while PLC protection stays independent; and an extensibility plane admits plugins, drivers, agent tools, and SDK clients through sanctioned registration services. Scenario knowledge sits in blueprints, signal maps, predicates, and driver configuration—not in team scheduler, approval service, or persistence code; members are interchangeable across 14 engine adapters under preserved identity, and 118 declared settings (40 hot-reloadable) plus the typed SDK make team topology, bindings, and limits data rather than code. In every plane the execution path is unchanged—a scenario supplies data and strategy (Fig. 1).

A. Supervisory Core and Nodes

The Node.js core hosts channel routing, industrial services, and protocol adapters. A channel contains a lead, workers, tasks, messages, and retrievable memory, deployed through WebSocket, MCP, A2A, REST, CLI, or TUI entry points; channel membership alone confers no write authority.

DAQ nodes convert device readings to engineering quantities across four modalities—scalars, vector profiles, image frames carrying content fingerprints, and mirrored manufacturing-execution-system (MES) records—so plant data-shape heterogeneity does not reach the team loop. MES endpoints are reached in two interchangeable modes—mirror, pulling MES records through the same acquisition pipeline as any field device, and direct, declaring read and write mappings against the MES interface. DCW nodes translate engineering setpoints into driver commands and, where supported, readback; per-node runtimes manage timing, and gateway services supply connection state, recipe context, and write bookkeeping. The acquisition surface declares six protocol families, five with drivers today—Modbus TCP, Modbus RTU-over-TCP, OPC UA, MQTT, HTTP/REST, and an S7 stub that fails loudly until its driver lands; the write surface is six drivers over the five writable families plus MES REST, with the evaluation's mock driver registered explicitly and outside that count; the evaluation instantiates five families as lines (Section V-C). The deployment reaches author-implemented dynamic simulators through the same node and driver contracts a plant would use; they are deterministic per seed, not validated digital twins. The first reuse boundary is declarative—signals, units, scaling, precision, and driver configuration—and the team loop is unchanged; fast regulation and independent protection stay in plant controllers.

Driver contracts are deliberately narrow: a driver declares how a semantic quantity maps to its transport payload, whether the endpoint is writable, and whether a readback is available, while the controller resolves member authority and active limits. Errors attach to the request rather than one success flag, and register addresses and byte order appear only here, so a new fieldbus family leaves the loop untouched. Undeclared drivers fail loudly: a node whose driver is not registered is rejected at creation and connection test rather than silently served by a mock. Acquisition is supervisory rather than hard real-time: each DAQ node runs its own cadence under a rotating sweep, so a slow endpoint delays only its own samples, and the

verdict stage queries the timestamped, quality-stamped history rather than issuing a blocking read.

B. Bindings, Context, and Reuse Boundaries

A member–node binding grants access to a sanctioned tool on a specific node and selects automatic or approval-mediated execution. Line permissions (none, read-only, operate) authorize binding creation and mutation; runtime invocation is authorized by the persisted capability, not channel membership. Intervals are re-checked every request instead of cached, so revoking a binding or switching the recipe takes effect on the next request; the same path serves a console, a harness, and an agent engine, because authority does not depend on the client—only the evidence record differs.

Production objects are versioned rather than overwritten: a recipe is dispatched against a known-good version, reverted non-destructively, and compared with the values actually applied, and an intervention is read against the definition that was active when it was admitted. Plugins register hooks, routes, agent tools, drivers, and panels only through sanctioned services, so an added driver inherits admission, readback, and evidence capture. Scenario reuse is bounded by an onboarding contract: the engineer declares a blueprint (devices, signals, units, roles, setpoints); a differential "ensure" pass creates missing devices, repairs drift, verifies connectivity, and projects it into semantic nodes and a recipe; a channel is built from the team template, and a mission supplies objective, guards, response windows, and policy. A BOPET worker proposes casting-speed, fast-roll, or rail-width changes through bound DCW nodes; the other three scenarios reuse the same pattern (Section V-F). The stores share identifiers, not a transaction domain: a device may accept a write even if bookkeeping fails.

C. Governed Optimization Task Life Cycle

A governed optimization task is a life cycle rather than a single tool call, and AgentWorkShop exposes each transition separately. The task enters a channel carrying goal, line, product, recipe, run, and task identifiers; a lead member claims it and hands it to a worker holding member–node bindings on the relevant nodes, whose fixed job then repeats: read the controlled quantity over a bounded window, propose a setpoint inside the resolved interval, invoke the governed write, wait for the response, and record a verdict (Section V-B instruments it).

Each iteration leaves more than a written value: agent- and rollback-source writes open an optimization record; the acknowledgment grade closes the command path; the journal attributes the change; attainment is computed against the scenario predicate while the task closes with a workflow terminal state (per-intervention detail in Section IV). Two write lifecycles coexist: direct node writes (agent dcw_control or admin REST) open these records, while recipe-chain operations (batch dispatch) are gated by the run ledger and open none (Section IV). This bounds

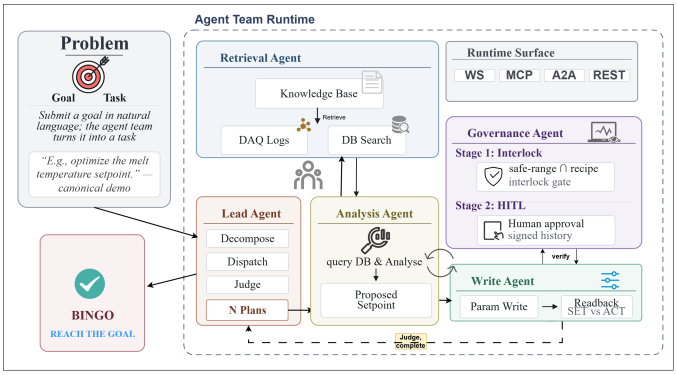


Fig. 2. Implemented agent-team runtime. A natural-language goal enters as a task; the lead agent decomposes and dispatches plans; retrieval and analysis produce the proposed setpoint (the LLM engine occupying a replaceable strategy slot). The governance stages are deterministic platform logic: Gate 1 applies the Eq. (1) interlock, Gate 2 the approval wait; the write agent reads back the value actually present (SET versus observed ACT), and the judge records keep, rollback, or uncertain; every step is journaled with actor attribution. The block labeled "Governance Agent" denotes these deterministic platform stages, not a model-bearing agent; the judge annotation is the caller's verdict-tool invocation.

what "automatic optimization" means here: a controller, a scripted policy, a conventional optimizer, or an agent engine may occupy the strategy slot, while authority, evidence, and closure stay on the platform.

IV. Governed Execution Mechanisms

A supervisory intervention must preserve distinctions that an agent tool call usually collapses—permission to propose, admission of a value, driver acceptance, device confirmation, process response, a caller's verdict, and compensation—and AgentWorkShop exposes these as separate evidence states. Three design commitments generate that separation: model output is treated as an untrusted proposal, never as authority; evidence is graded by what is observed rather than by what was claimed; and failure states are surfaced for inspection rather than absorbed into a success flag. Fig. 2 shows the implemented agent-team runtime that carries these steps; the governance and write stages drawn in the figure are deterministic platform logic—no model output can admit, approve, or confirm a write (Gate 1: admission interlock; Gate 2: approval wait). A governed intervention is one transaction in the audit-trail sense: every step leaves its own evidence record, while execution and recovery are deliberately not atomic with device state (Section VI). Six invariants organize the path: authority resolves from the member–node binding, never from channel membership or prompt content; the proposal must lie in the Eq. (1) interval; a manual binding inserts a timed, fail-closed approval wait; the write leaves a graded acknowledgment, not a success flag; three-way setpoint consistency can be checked before any verdict; and compensation is a separately admitted write referencing the record it reverses. Every agent-originated path that AgentWorkShop exposes to the plant passes through the same governed write.

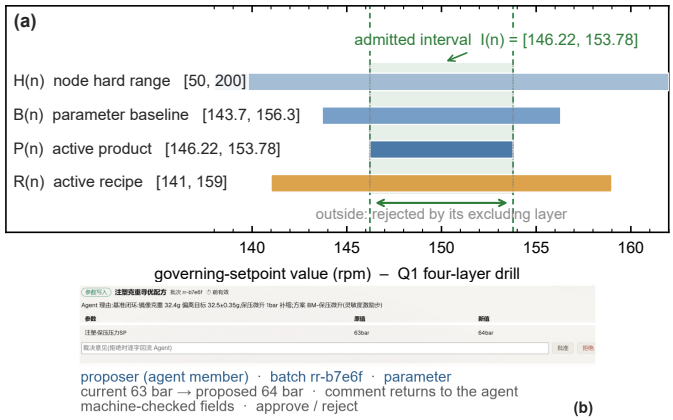


Fig. 3. The admission boundary and its human-approval gate, on the running benchmark line. (a) The Eq. (1) interval geometry: four declared, versioned bounds narrow to the admitted interval; a proposal outside it is rejected by its excluding layer (the Q1 four-layer drill). (b) The structured approval card on a manual binding (live system, gloss added): machine-checkable fields are separated from the proposer's display-only reasoning; the comment returns to the member on refusal (Section IV-E).

A. Task, Observation, and Admission

The strategy receives a task and goal in production context and decides what to propose and when to stop; the platform governs execution over semantic node parameters—pressure, dosing rate, line speed, rail width—in engineering units.

For an ordinary agent setpoint request on node n, the admission interval is

$$\mathcal{I}_n(t) = \mathcal{H}_n \cap \mathcal{B}_n \cap \mathcal{P}_n(t) \cap \mathcal{R}_n(t), \tag{1}$$

where $\mathcal{H}_n$ is the mandatory hard engineering range (finite and non-empty at node creation) and the remaining terms the baseline, active-product, and active-recipe limits; a missing optional limit contributes no restriction, so the interval is always defined, and t is the admission instant, re-resolved at execution. Should a configuration empty the intersection, every write is rejected with the excluding layer named, so the misconfiguration surfaces at the first write. The interval form is deliberate: every bound is a declared, versioned object that an audit can replay against the active product and recipe, and rejection is decidable and layer-attributable, where a learned feasibility model offers neither (Fig. 3a). The claim is decidability and replayability, not learned-model safety. It bounds one parameter at a time; joint, cross-variable windows live in scenario guards. It is not a physical safety certificate, and bounds magnitude, not intent: availability, binding state, and approval policy are checked separately, and in-window but adverse sequences surface at judgment and compensation (Section V-F reports two).

B. Semantic Write, Pacing, and Source-Specific Routes

A semantic card exposes quantity, unit, precision, range, value, freshness, and recipe context; the controller resolves binding and limits, the driver serializes the command, and the acknowledgment grade—not driver success alone—closes the command path (Section IV-C), never the process-response path. Two write-spacing windows pace governed loops, never the plant: online writes—agent- or human-origin—sit at least 60 s apart with a hold window after success (HTTP 429 inside it); a per-recipe dispatch interval anchored at the approval instant is restart-persistent, so alternating identities cannot reset it; and a 300 s per-line trial cadence damps same-line trials from any origin. Recipe-dispatch and rollback writes are exempt. AW_BENCH_MODE=1 admits harness-declared arms that relax the soft window or readback—never the hard range.

Four governed routes—agent tools, human-origin writes, recipe dispatch, and rollback compensation—reach the same driver through one admission and journaling entry point, the hard range included; recipe dispatch adds run context—the line-scoped execution window, the reduced ISA-88 batch/lot notion—validated at save time. Hold heartbeats (periodic re-dispatch of the stored setpoint) are the one non-governed re-dispatch: off by default, stopped by line stop or driver change, and re-applying the stored value without re-admission—plant-facing state maintenance, not an agent path. Line start installs active context before recipe parameters dispatch individually, and line stop clears it; these multi-step operations are not atomic with device state.

C. Write Acknowledgment as Three-Level Evidence

A driver-level success flag conflates two facts: the transport accepted the command, and the device now holds the value. AgentWorkShop grades every governed write into one of three acknowledgment levels: readback-verified when an independent read—one issued outside the write path—returns the commanded value within the node's configured write tolerance; transport-acknowledged when the transport accepted the command on an endpoint with no usable readback; unverified when the transport failed or when post-write verification could not confirm the value. The grading rule is falsifiable. Under a success-flag receipt rule—a transport acknowledgment closes the report—the publish-only receipt demonstrated in Section V-G (broker puback, ok=true, no read channel, three verification attempts) reports success; the graded rule reports it unverified. That is what fail-honest means: absence of evidence degrades the grade, never the report, and the grade is fixed when the write closes.

A transport-acknowledged write may still not have reached the device, so a post-write verifier re-reads the endpoint (three attempts, 2 s apart) and promotes the grade when a read confirms the value. Batch receipts carry a per-parameter grade plus a summary; the write journal stores the per-write grade, and the operations log records the judgment at worst-state severity. Recipe rollback restores a definition, not physical PLC values, and a compensating write is itself graded by the same three levels. The grade closes the command path only: readback-verified says the device holds the value, not that the

process responded as intended—that distinction belongs to observation and verdict.

D. Observation, Judgment, and Compensation

An agent- or rollback-source write opens an optimization record; the verdict tool dcw_judge records keep, rollback, or uncertain plus a reason; the verdict, stored with the record it closes and the response window actually queried, performs no actuation and does not independently validate the evidence, so a scripted keep can coexist with a missed objective—verdict integrity rests on attribution, journaling, and operator review: a kept-but-marked record stays visible with its disagreement attached, and the operations log raises parameter severity under alarm-rationalization practice (Section VI). Three-way setpoint consistency is checked by the evaluation harness rather than the platform: the harness compares the recipe value, the device readback, and a mirror sample drawn from the acquisition runtime's separate polling path and store, each leg against a stated tolerance; the sampled legs meet at the plant endpoint, so a fault fabricating the endpoint is a common-mode boundary. Disagreement fails the harness check and is reported with the run. Verdicts are attributable—actor and timestamp travel with them; corrections amend the stored reason visibly, and compensation opens a new record referencing the one it reverses.

Compensation is a new, separately admitted write that may fail or partially complete; it is graded by the Section IV-C levels, and a rollback record settles to rolled back once the compensating write is accepted—readback-verified where the endpoint supports readback, transport-acknowledged otherwise. If the compensating write is itself refused by admission, the source record closes as judged, with the refusal appended to its verdict. These separations keep command status, record status, and process recovery independently inspectable.

The backstop is a record-integrity check for writes that no verdict has closed—not an independent protection layer: it evaluates only open records; a caller's keep closes the record and stands it down, while a rollback verdict stands the backstop down only once the compensating write executes—so its availability is contingent on verdicts being recorded and observations arriving. When collected samples of an open record's node leave the recipe's monitoring window (a DAQ-side envelope distinct from the admitting interval), the platform issues a system-attributed rollback verdict and restores the value preceding that write (one step; the admitted baseline is not restored), graded as above. The policy waits 120 s, polls every 30 s, and fires on three out-of-window samples—cumulatively, not necessarily consecutively; non-returning samples count no breach, so an acquisition outage leaves the record open and unalarmed. Recovery runs only while fewer than two retained rollback records stand on the node, past which the backstop judges but defers actuation to the governing approval policy.

E. Human Approval Boundary and the Line Master Switch

Manual bindings insert the timed approval wait before an agent setpoint or compensation request; automatic bindings omit the wait but keep the checks. The pending request shows member, node, value, interval, and expiry—in the deployed benchmark, plus the proposer's reasoning and per-parameter basis (Fig. 3b): machine-checked fields are structured data, the free-text reasoning display-only and non-authoritative. Because binding and admission are re-checked at execution, approval reserves neither interval nor device state; the approval dialog—itself an injection surface (Section VI)— cannot promote a proposal past the checks; the comment returns verbatim to the proposing member on refusal and travels with the receipt on approval—feedback, not authority. The wait fails closed by default (expiry 180 s): an expired request resolves as refusal and is recorded; duplicates are suppressed by holding one pending request per member–node pair; and in hold mode an unanswerable request persists without expiry until a human answers, with repeat requests refused as duplicates.

Human authority is anchored one level above individual requests: each line carries a control mode—the line master switch—whose fail-safe default is manual. Under manual mode, recipe-family operations (parameter update, dispatch, trial, and rollback) from approval-free bindings on that line are intercepted and presented for approval regardless of binding policy; direct node-tool writes remain gated by their binding mode and the admission interval, not by the line switch. Switching a line to automatic requires an explicit, admin-only user-side confirmation—agent identities cannot touch the line switch. The residual agent-side path is delegation: a lead agent can grant held nodes to a worker through an attributed lead-only tool, and the grant's execution mode is chosen at grant time rather than gated by the user-side confirmation; grants are attributed to the granting agent, cannot exceed its own node face, and are individually revocable (Section VI).

The decision endpoint authenticates a user but does not enforce a line-scoped approver role—the gap Section VI carries—and reads, verdict recording, and some configuration changes do not use physical-write approval—workflow mediation, not segregation of duties; a recipe switch between approval and execution re-admits the value against the new window (Section VI). The strategy engine is replaceable rather than trusted as a control primitive: the stable plane keeps bindings, semantic I/O, evidence, approval, compensation, and backstop handling, while scenario code declares and configures—never executing plant I/O—and the platform never infers a process objective.

V. Experimental Evaluation

The evaluation asks whether one governed supervisory path can be exercised, inspected, and repeated across heterogeneous industrial jobs, not whether one optimizer beats another. Five questions organize the evidence: protocol closure and admission-gate attribution (Q1), the

AgentTeam workflow (Q2), a deterministic controller loop (Q3), four concurrent missions (Q4), and deployment-scale regression with write-evidence verification (Q5). Q2–Q4 occupy the strategy slot (scripted policies, with two live engine turns in Q2); Q5 stresses the hardened platform under load. Together the five exercise each production-facing plane of Section III.

A. Testbed, Evidence Snapshot, and Scope

The evaluation draws on two evidence tiers of one codebase. The archived snapshot for Q1–Q4 is integrated run 20260921184204-10lc (seed 42, commit e692df1, harness 9f2756dce4a4cbb3, completed September 21, 2026, 19:08 UTC), archived with raw reports, plus ablation companion 20260921181234-e1lite. Its standing preset is a cast-film extrusion line tuned through the die-head heater setpoint (objective J weights thickness conformance, defects, energy, and throughput under melt-temperature and pressure constraints, larger better). Each mission contributes one seed-42 trajectory: Q1, Q2, and Q4 are single-trace case studies (every claim each makes is checkable in the archived artifacts), the deterministic loop adds two in-run traces (seeds 43–44 of the same campaign), and repeat evidence comes from Q5's consecutive deployment-tier campaigns—a different tier, not a re-run of these traces.

The deployment-scale tier for Q5 is a live v0.7.57 instance carrying 34 provisioned lines, 112 agent channels, 67 online DAQ nodes, 77 write nodes, 52 recipes, and 385 member–node bindings—simulator- and bench-backed objects, not physical plants; multi-line means many provisioned lines and channels on one host. The benchmark line is a physics-based injection-molding cell (16 DAQ nodes over the four modalities, 6 control nodes over five protocol families plus MES); all Q5 numbers come from committed per-run reports, ledgers, and timelines (October 7–8: hardened-pipeline run -20261008045434 plus three consecutive full-suite runs; October 9: benchmark-20261009021149/-024141/-033810); mechanisms are unchanged since the Q1–Q4 snapshot except the October hardening pass's acknowledgment grading and master switch; the campaign ran under live load. The harness explicitly sets the benchmark line's master switch to automatic with a user-side confirmation—the automated path is the object under test—and verifies the switch's manual-mode interception separately (Section V-G, leg 1).

B. Test Procedure and Campaign Overview

The Q1–Q4 procedure is a 20-phase integrated suite—an author-defined regression instrument, not an external certification—probing one claim per phase, from preset loading against the best grid point ($J^* = 89.894$) through protocol interception and the record, recipe, approval, and parameter-layer drills (Sections V-C and V-F).

The Q5 procedure is the one-command benchmark pipeline: eight stages (S0 pre-check; S1 API surface matrix, a positive-and-negative sweep across authentication, users, channels, tasks, permissions, both node families, agent tools, HITL, logs, memory, plugins, teams, and notifications; S2 multi-modal ingestion with MES dual-mode; S3 governance negative paths; S4 closed-loop optimization; S5 fine-tuning; S6 diagnostics; S7 report) under a harness that captures every check and a timestamped event timeline per run, so each stage exercises one functional surface of Sections III–IV. The profile has since grown (S1 adds login lockout, S4 the acknowledgment-grade summary); the October 9 campaign re-runs it three times. No prior system in Table I owns a governed write path to plant endpoints, so the primary comparator is internal, and two controls are archived. The admission control holds the write path fixed and relaxes its mechanisms: the six window-edge probes that full governance rejects (HTTP 400) return 200 executed under the relaxed arm (Section V-C, 18/18→12/18)—a success-flag design reports them as accepted writes. The evidence control holds the receipts and varies the reporting rule (Section IV-C); the success-flag outcome is definitional—the pre-hardening harness itself reported a broker puback as success. The closed-loop harness also parameterizes the write injection (admin REST beside the agent's governed dcw_control); a like-for-like pair on the shared live instance (runs -20261012042856, -043509, -051224, -060026) was rejected in full on both routes—24/24 writes per arm under production pacing, and 24/24 again with pacing exempted; the archived runs record aggregate rejections only, and a closed-loop quality comparison awaits a dedicated instance. The harness's own reports verdict these attempts fail; they are archived as gate evidence. Fig. 4 summarizes the closed-loop instrument.

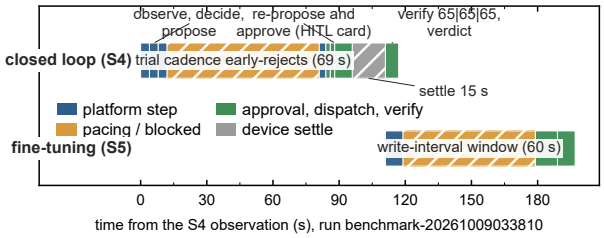


Fig. 4. One governed iteration on the October 9 line (harness-adjudicated approval): the decision is corrective (0.276 g error) while the proposer packages the step as a sensitivity excitation, and the trial cadence early-rejects 69 s before a human is involved; structured approval releases the re-proposal, dispatch returns readback-verified, and the three-way check reads 65|65|65. Lower lane: fine-tuning under the same 60 s window, rolled back and restored.

C. Q1: Protocol Closure and Admission-Gate Attribution

On the five protocol lines the suite stores 55 DAQ samples through real drivers and dispatches governed writes over four protocol families—about 6 writes each, p50/p95: 47.8 ms/248.9 ms (Modbus TCP), 19.8 ms/20.5 ms (OPC UA), 19.3 ms/20.1 ms (MQTT), 32.9 ms/44.6 ms (HTTP); the RTU-over-TCP satellite is acquisition-only—recording readback confirmation on the endpoints that support it, with one admission,

TABLE II
Admission ablation on companion run 20260921181234-e1lite: six probe classes per arm, three deterministic repetitions. The four hard-range classes are rejected in every arm; only the two window-edge classes flip.

Arm	Soft win.	Readback	Intercept	Executed
full	✓	✓	18/18	0
no-readback	✓	—	18/18	0
no-interlock	—	✓	12/18	6 jour.
ungated	—	—	12/18	6 jour.

record, and verdict path serving every request. Each writable line rejects 6/6 out-of-window probes (24/24 intercepted) while accepting its legal write; the window-interlock drill probes the window edge and both admission branches. Two drills separate truth paths one success flag would collapse: a frozen-measurement drill (write and readback consistent, measurement out of window) ends with a system-judged backstop rollback 130.2 s later, and a record-rollback drill restores the prior value by a separately admitted, readback-confirmed write. A second preset with different signals and dynamics is commissioned by configuration alone; across both presets 33 probes are rejected on seven writable lines, the same admission code discriminating on unseen signal maps. The parameter layer makes the interval geometry of Eq. (1) observable (Fig. 3a): a four-layer chain—node $[50, 200]$, parameter $[143.7, 156.3]$, product $[146.22, 153.78]$, and recipe $[141, 159]$ rpm—narrowing to one admitted interval, each rejection attributable to its excluding layer; a manual binding suspends a request visibly and unblocks it on approval (latency 461 ms, harness-adjudicated and poll-dominated; Q5 reports the server-side span alone).

The ablation replays the same probes on companion run 20260921181234-e1lite and a standalone fixture (six probe classes per arm, three deterministic repetitions with identical counts); it isolates the window interlock, untouched by the October hardening pass. Table II reports the four arms: full governance and the no-readback arm reject 18/18; without the window interlock (alone or with the readback check) only 12/18 reject and the six window-class probes execute and are journaled—prevention is the interlock's job; a relaxed configuration degrades from prevention to detection, with the readback check retained to make written-but-unconfirmed commands visible. The hard range is structurally unbypassable—all four governed routes re-admit it at one journaling entry point, per request (Section IV-B)—and the soft window is its removable part.

D. Q2: The AgentTeam Optimization Workflow and Its Results

The workflow ran end to end on the cast-film line and then on a larger BOPET line (Section III-B); a scripted policy rather than a language model issued the writes (two live engine turns follow). On the single-node mission one write sufficed ($200 \rightarrow 204.704\,^{\circ}$C, settled at $204.70\,^{\circ}$C, journal-attributed, task COMPLETED). The BOPET mission is the multi-node case: nine devices and 49 signals (30 writable, 19 readable), all ensure-pass created. Three governed writes on three distinct nodes—casting speed $32 \rightarrow 33.8$ m/min, fast-roll speed $118 \rightarrow 120.3$ m/min, and rail width $3000 \rightarrow 3035$ mm— bring thickness from 28.10 to 25.66 μm against $25.0 \pm 0.7\,\mu$m—0.04 μm inside the window; the same run's prose summary records it not attained while its machine-readable verdict records attained, so we report both. Each write opened a record, received keep, and left an agent-attributed journal entry, the parent task reaching COMPLETED; governed-write counts match accepted changes in both missions. Finally, two live agent-engine turns (omp adapter, glm-5.3-flash) drove the same slot on a scratch line—a bounded write (156.3 s) and a goal-driven multi-step descent settling at 51.7 μm against a harness-supplied target ($52.0 \pm 0.8\,\mu$m written into the task text, not the scenario predicate), saving a recipe with attributed, audit-logged authorship (637.3 s). They show that an engine can occupy the slot, not agent quality: the run's pass criterion is a completion marker in the transcript.

E. Q3: The Deterministic Closed Loop

Seeds 42–44 each issue two governed writes with no rejection, raising the cast-film objective J from $J_0 \approx 67.2$ to $J/J^* = 0.965$–0.972, against the best grid point $J^* = 89.894$—an exhaustive grid search over the admissible setpoints (zone temperature 195–225 °C in 2.5 °C steps, screw 50–200 in steps of five, line speed 20–120 in steps of 2.5 m/min) on the same simulator, so J^* is a lower bound on the true optimum and the reported J/J^* is an upper bound on J/J_{true}: the true gap is at least as large as shown. What the benchmark establishes is narrower: the governed path passes a conventional controller's writes through unmodified—admissible, journaled, read back. The claim is transparency of evidence handling, not a performance delta.

F. Q4: Four Process Missions

The campaign runs four blueprint-provisioned lines concurrently; in this run (seed 42) all four attain their archived predicates—evaluated from subsequent observations, not task state. Fig. 5 archives each mission's optimization trajectory. Injection molding is the minimal case (two writes: hold pressure 45→58.2 bar, hold time 6→10.64 s), BOPET the multi-node case of Section V-D; both close with task state and predicate agreeing, on separate records. Wastewater treatment is the write-heaviest mission (15 governed writes): dissolved oxygen rises from 0.21 into 3.4 ± 0.6 mg/L (attainment window; terminal observation 3.50 mg/L, Fig. 5b) under five effluent guards: four guard margins finish wider than at baseline, and the fifth, pH, drifts 6.59→6.38. Its most informative events are blower steps that were admissible and journaled yet lost compliance margin in the post-write observation; the verdict stage reverted them, so a bad proposal is rejected in the evidence, not at the admission gate. Annealing starts far outside the hardness window (134.52 HV at

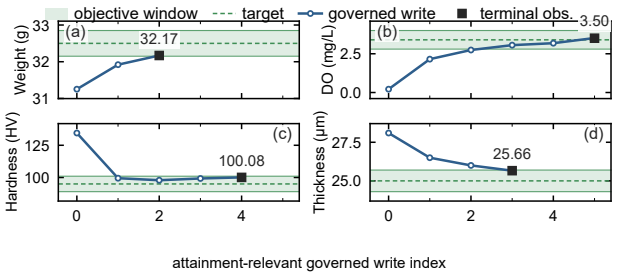


Fig. 5. Archived optimization trajectories of the four concurrent missions (run 20260921184204-10lc, seed 42): observed controlled quantity after each attainment-relevant governed write, against the archived objective window; the index counts those writes, so the 6 plotted points of (b) are 6 of its 15 governed writes; (a) injection molding, (b) wastewater treatment (A2O), (c) annealing, (d) BOPET film. DO, dissolved oxygen. Simulator observations under scripted policies.

140 m/min); zone-3 temperature, one zone-2 trim, and two line-speed steps hold 95 ± 6 HV while raising line speed by 17.1%, and a further attempt at 176 m/min is reverted for leaving the margin—on record. Mission clocks (26.48 s, 169.27 s, 102.76 s; BOPET's unarchived) are simulator wall time under accelerated dynamics. Across the four missions in this run, platform-side behavior is identical while the write economy—governed writes per scripted policy—varies by more than a factor of seven (2, 15, 5, and 3).

G. Q5: Deployment-Scale Regression and Write-Evidence Campaign

The final lens stresses the platform itself: the hardening pass made the evidence grading, the line master switch, and the approval-anchored dispatch interval of Section IV the explicit test surface of the closed-loop, fine-tuning, and write-evidence scenarios on the live instance of Section V-A. Full-suite repeatability. Seven pipeline executions complete with zero failed checks—79 checks per run on October 7 (237 total), 91 on the hardened-pipeline run, 90–92 per run on October 9 (600 total); two runs land inside the per-line trial cadence (Section IV-B) and wait out the remaining 218 s— pacing stays in force under automation.

Closed-loop scenario under governance. Each run walks observation, grey-box-sensitized decision (dead-band 0.15 g, slope 0.052 g/bar), a five-element per-parameter proposal, HITL approval, batch dispatch, and three-way consistency (recipe, PLC readback, mirror sample). The loop runs in two regimes: October 7–8 descends 66 → 63 bar one governed step per run under converged dead-band excitation; October 9 answers a sagging mirror weight (32.49/32.33/32.22 g) with a bounded zigzag (64→63 bar, converged) and corrective steps (63→65 bar). The zigzag is a deliberate excitation, not a correction—the archived decision records the loop already converged (0.014 g against a 0.15 g dead-band) and names exercising the dispatch chain as its purpose—while the two corrective steps answer residual error (0.169/0.276 g). Every step verifies equal values across the three sources at maximum consistency error 0.0 bar (0.05 bar device, 0.51 bar bucketed-mirror tolerances—all three legs equal exactly at the endpoint's recording resolution, a passed tight check, not equality by construction). The pacing window acts before the human: in two runs the proposal is early-rejected by the trial cadence and re-proposed 69 s later; only then is the HITL card raised (scheme, proposer reasoning, per-parameter basis) and structured approval releases the readback-verified dispatch (Fig. 4). Approval-to-dispatch completes in 6–9 ms; observation-to-verification spans 25.1 s without a pending cadence window (245.20/245.18 s with, October 7–8; 96.1 s, October 9)—the loop's time is governance, not platform overhead. The fine-tuning scenario rides the same windows (Fig. 4, lower lane), and the diagnostics scenario exports a 30-min merged wide table (663–869 rows, 14 columns)—a feed data-quality observation, not a process excursion.

Latency and mission figures are single-run point estimates (about 6 writes per line, one trajectory each); only Q3 and Q5 are repeated.

Write-evidence legs. A dedicated seven-leg suite verifies Sections IV-C and IV-E directly: (1) the master switch refuses an unconfirmed manual-to-automatic transition; (2) a mock-acknowledged dispatch is promoted to readback-verified; (3) a broker cut marks affected parameters unverified while healthy siblings verify; (4) a publish-only endpoint accepts the write (broker puback, ok=true) yet stays unverified through three verification attempts—the receipt Section IV-C's falsifiable rule is stated against; (5) on a manual-mode line an approval-free binding's dispatch is intercepted into the queue, approval releasing all six parameters device-verified; (6)–(7) API regression and documentation-consistency suites close the campaign.

Across the five questions the pattern holds: admission intercepts and attributes, grading degrades honestly, pacing holds under automation, approval mediates without carrying the checks, and compensation and the backstop recover on record—Section IV's mechanisms behave as specified, with task state and attainment never merged into one number.

VI. Discussion and Limitations

A. Industrial-Informatics Interpretation

One supervisory path carried four different jobs and a deployment-scale campaign with command closure, verdict, task state, and attainment kept as separate records. Against the nearest systems of Section II, the differentiator is not the LLM—a replaceable strategy slot—but the production-scoped authority and evidence around its writes: no nearest system reports that combination on one path. Deployment-wise, the layer installs as one service beside the plant's protocol endpoints: a line is onboarded by declaring a blueprint, and approval clients—web panel or API—receive the HITL cards; PLC protection and DCS interlocks stay untouched.

B. Validity and Reproducibility

Run identifier, seed, commit, harness fingerprint, and raw artifacts are fixed, so counts are re-derivable rather

than trusted; Q5 adds per-run ledgers and timelines. Five qualifications bound what they support: simulation-tier evidence (deterministic per seed, not validated digital twins; no physical-device campaign, operator study, or external comparison reported); internal-oracle optima and predicates; single-host latencies and near-contemporary runs; implemented-but-unexercised capabilities and a single live provider/model; and harness-adjudicated approval (human decision latency and quality out of scope).

C. Safety Positioning and Human Oversight

The threat model is a misbehaving or manipulated agent (wrong or hallucinated proposals) inside an otherwise trusted platform and plant; the approval dialog, admission re-checks, and journaling are the containment surfaces, and no adversarial injection campaign or operator study of the approval dialog is reported; stolen credentials and compromised endpoints are the domain of IEC 62443 [18]. Fast regulation, hardware interlocks, and safety functions remain in the PLC; prompts and retrieved documents are untrusted text, not authority [25], [26]—the proposer is confined to per-binding scope and journaled, attributable actions: a bounded blast radius, not a protection guarantee. Admission limits, acknowledgment grading, and the backstop run with or without human approval—enforcement never queues behind human attention. The pacing windows bound approval demand per line (a write, a trial, or a batch release no oftener than the windows allow, and expiries resolve as refusals), and batch release itself—one approval covering a six-parameter dispatch (Section V-G, leg 5)—is the demonstrated fatigue mitigation; approval fatigue and automation bias remain human-factors risks nonetheless, and reading graded evidence assumes trained approvers [24], [27]. A lead agent's delegation-mode choice (Section IV-E) is the one agent-reachable path around per-request approval; capping granted mode at the granter's own binding mode, or requiring the line-switch confirmation for it, is the queued guard.

D. Deployment Boundaries

Execution and recovery are non-atomic: bookkeeping may fail after actuation, the pacing windows time writes rather than certify stability, declared per-write step limits are advisory metadata surfaced to agents rather than enforced bounds, readback agreement and three-way consistency need not imply process recovery, and multi-node compensation may partially succeed; hold heartbeats re-apply the stored setpoint without re-admission and can trail a narrowed window; oscillation, storage failure, clock error, and concurrency beyond the four-line campaign remain unevaluated; redundant supervisor deployments with divergent stores (split-brain) are unevaluated; a supervisor outage halts dispatch rather than permitting writes through, and a hardware-in-the-loop leg against a real PLC or OPC-UA endpoint is the next evidence tier. Production integration additionally requires: bound changes managed under change-control with process-evidence validation; approval-aging, open-record-aging, and backstop-deferred states surfaced on the plant alarm system; the bench-mode environment switch refused at production boot; and approval treated as workflow mediation—attribution and timestamp, not an electronic signature. In production, bindings govern sanctioned tools, not arbitrary privileged-engine I/O—least privilege, endpoint security, and IT/OT separation remain necessary [18]; AgentWorkShop is neither a safety instrumented function nor a plant-protection substitute [28]; line-scoped approver authorization is still missing, so production needs segregation of duties and alarm-path failure tests [23].

VII. Conclusion

AgentWorkShop contributes a governed supervisory execution architecture rather than a new process optimizer: one stable path from agent-team identities through semantic nodes, production context, governed writes, graded acknowledgments, verdicts, and compensation, exercised by a 20-phase suite, agent-team missions, a closed loop, four concurrent missions, and consecutive deployment-scale regression campaigns on a live multi-line instance. Physical-plant deployment, operator studies, and external-framework comparison remain open; until then, optimization benefit, autonomous control, and safety assurance stay unsupported claims.

Data Availability and Disclosure

Artifacts are archived per run under bench/results/ and docs/benchmarks/; anonymized artifacts are available during review and released with the code upon publication. The two live engine-turn console transcripts (Section V-D) were excluded by the repository's log-retention rule and survive only as recorded outcomes; the write-evidence suite's checks are documented in the audit report and rerunnable from its script. AI assistants supported editing under author direction; the authors remain responsible for all claims.

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